Olist E-commerce SQL Project

Business Problems and Analysis Questions

Background

Olist is a Brazilian online marketplace that connects small sellers with customers across the country. Order volume has grown quickly, and leadership wants to know whether that growth is healthy.

Business Problems

1. **Revenue concentration:** Where does revenue come from, and does it depend too heavily on a few categories, states, or sellers?
2. **Delivery and satisfaction:** Are late deliveries hurting customer review scores, and which regions have the worst delivery performance?
3. **Customer retention:** Are customers coming back, or is growth driven only by new buyers?

Goal: Use SQL to analyze about 9,300 orders ([period, e.g. January to December 2017]) and recommend where to focus to grow revenue, improve delivery, and keep customers.

Analysis Questions

Problem 1: Revenue concentration

1. What is the monthly revenue and order count, and how fast is it growing?
2. Which product categories earn the most revenue?
3. Which states earn the most revenue, and what share do the top 3 make up?
4. Who are the top 10 sellers, and what share of revenue do they and the top 10% of sellers make up?
5. What is the average order value, and which payment methods and installment levels are linked to larger orders?

Problem 2: Delivery and satisfaction

6. What percentage of orders arrive later than the estimated delivery date?
7. What is the average delivery time, overall and by state?
8. Which states have the highest late delivery rates?
9. How do average review scores compare between on-time and late orders?
10. How does the review score change as the delay gets longer (1 to 3 days, 4 to 7 days, 8 to 14 days, 15+ days)?

Problem 3: Customer retention

11. How many unique customers are there, and what percentage ordered more than once?
12. How many days pass between a customer's first and second order?
13. Which states and categories have the highest repeat purchase rates?